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SEISMO-VRE: A tool for a multiparametric and multidisciplinary study of an earthquake

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ABSTRACT

The study of earthquake preparation phases often relies on fragmented approaches, limiting reproducibility and comparison between methods. To address this, we developed a Virtual Research Environment (VRE) for multiparametric and multidisciplinary earthquake investigations. Built as a Jupyter Notebook with MATLAB and Python kernels, the VRE integrates seismic, geodetic, atmospheric, and ionospheric data into a unified and automated workflow. Users can define spatial, temporal and other parameters to retrieve and process data across layers. Its effectiveness is demonstrated through the analysis of the 2016 Central Italy and 2025 Marmara earthquakes, where the tool proved capability to easy reproduce cross-domain results.

Metadata

Nr	Code metadata description	*Metadata*
C1	Current code version	*v1.0*
C2	Permanent link to code/ repository used for this code version	https://github.com/dedalomarchetti/SEISMO-VRE
C3	Permanent link to reproducible capsule	https://mybinder.org/v2/gh/dedalomarchetti/SEISMO-VRE/cd8dbaab61c074d8b4e6392459fc7ec5840e0c3c?urlpath=lab%2Ftree%2FSEISMO-VRE_Python_v1.0.ipynb
C4	Legal code license	*GPL 3.0 License.*
C5	Code versioning system used	*git*
C6	Software code languages, tools and services used	*Matlab and Python under a Jupyter Notebook*
C7	Compilation requirements, operating environments and dependencies	*Any environment running JupyterHub or JupyterLab servers with MATLAB or Python kernels (specific dependencies are listed within the specific Notebook)*
C8	If available, link to developer documentation/manual	*The documentation is provided directly inside the Jupyter Notebook*
C9	Support email for questions	dedalo.marchetti@ingv.it

1. Motivation and significance

Understanding the preparatory phases of earthquakes requires integrating data from multiple geophysical domains, including seismology, ground deformation, atmospheric variability, and ionospheric observations. In recent years, various case studies have demonstrated the potential of such multiparametric and multidisciplinary approaches, including the analyses of relevant seismic events, like the 2015 Nepal (Mw = 7.8 and Mw = 7.3) [1], 2018 Indonesia (Mw = 7.5) [2], 2019 Ridgecrest (Mw = 7.1) [3] and 2020 Samos (Greece, Mw = 6.0) [4] earthquakes. However, these studies were typically based on bespoke, non-standardised workflows, developed ad hoc by individual research groups. This has limited reproducibility, slowed comparative analysis across events, and hindered the broader adoption of such approaches.

To address these issues, we developed an open-source Virtual Research Environment (VRE) that provides a reproducible and modular framework for the joint analysis of seismic, GNSS, atmospheric, and ionospheric data. The software is implemented as a Jupyter Notebook [5], available in both MATLAB and Python kernel versions, and is designed to be interoperable with research infrastructures such as the European Plate Observing System (EPOS) [6]. It allows users to define key earthquake parameters (location, magnitude, time, etc.) and retrieve the relevant data from remote sources via web services (HTTP APIs, FTP) or from local files. The VRE supports multiple usage modes:

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real-time streaming, full data download, and offline analysis, depending on the dataset size and availability.

The current configuration utilises reference datasets from EPOS Platform [7] (Thematic Core Services - TCS - Seismology [8,9] and GNSS [10,11]), NASA MERRA-2 [12–17], and ESA Swarm [18], which serve both as scientific baselines and as templates for future extensions. Researchers can easily modify or add data sources, making the tool adaptable to new case studies or even to different geohazards.

By facilitating standardised and reusable analysis workflows, the proposed VRE reduces the technical barrier to adopt a multidisciplinary approach for studying the earthquake preparatory phase. It also contributes to the broader goals of open science, data-driven discovery, and cross-domain collaboration.

2. Software description

This tool was developed in a Jupyter Notebook that can run both on a MATLAB and Python kernel, according to the user's favourite programming language. As JupyterLab can be executed under different Operating Systems (OS), the presented tool addresses portability requirements, thus improving reproducibility and reuse. Furthermore, as Jupyter Notebook can run on a server, the execution of the tool can also be performed remotely. Given the widespread adoption of Jupyter Notebooks in the scientific community—particularly within Research Infrastructures that provide data access and interactive Python-based environments—the proposed tool can be seamlessly integrated into platforms such as EPOS [7] as a multidisciplinary case study, leveraging their remote computation capabilities. Regarding the Python version of the Jupyter notebook, it requires some libraries that are described in the "Requirements" technical file. MATLAB version relies on system functions, and a few more functions are distributed along with the main tool.

2.1. Software architecture

The proposed software is structured to support the multiparametric investigation of earthquakes through the analysis of three main geophysical layers: the lithosphere, atmosphere, and ionosphere. This scientific structuring is reflected in the software architecture, which organises the workflow into independent, reusable blocks for data retrieval, preprocessing, analysis, and visualisation (Fig. 1).

The software is provided in two versions, MATLAB and Python, in order to satisfy diverse requirements from different scientific communities, thus maximising the impact, diffusion, increasing its usefulness.

The MATLAB version requires a standalone installation of MATLAB version 2020b or later, along with the module that integrates it with Jupyter Notebooks. Instructions and installation modules are available on https://github.com/mathworks/jupyter-matlab-proxy [accessed on 2 September 2025] and provided directly by MathWorks. 'Mapping', 'Statistics and Machine Learning' and 'Curve Fitting' toolboxes are required for specific functionalities. A "GNU Octave" version of SEISMO-VRE is under developpement and it will be provided as a dedicated notebook.

The Python version runs on Python ≥ 3 and requires libraries such as `numpy`, `pandas`, `matplotlib` and `scipy`. All dependencies are listed in the `requirements.txt` file. The software is fully compatible with cross-platform environments and can run on local machines, as well as on remote infrastructure via JupyterHub or JupyterLab servers [19].

In order to ensure environment-level reproducibility, the user can run SEISMO-VRE within a containerised execution environment provided through the Binder service (https://mybinder.org/v2/gh/dedalomarchetti/SEISMO-VRE/cd8dbaab61c074d8b4e6392459fc7ec5840e0c3c?urlpath=lab%2Ftree%2FSEISMO-VRE_Python_v1.0.ipynb). Binder contains a full copy of the repository with the required Python environment with all the necessary dependencies. Consequently, the user can just run SEISMO-VRE by opening the Binder link (included and updated also in GitHub) with default or own parameters.

Data access is performed through standard web services, including HTTP APIs (e.g. for seismic catalogue and GNSS data), FTP/HTTPS servers (when users download data manually, in the MATLAB version), and local file inputs. This ensures flexibility in choosing between different methods for accessing data. In addition, data can be accessed in the workflow both as real-time-downloaded data or as offline analysis using pre-downloaded datasets. At this purpose the user can pre-download the required large dataset by dedicated Jupyter Notebook available in "data" folder named "Download_data_for_SEISMO-

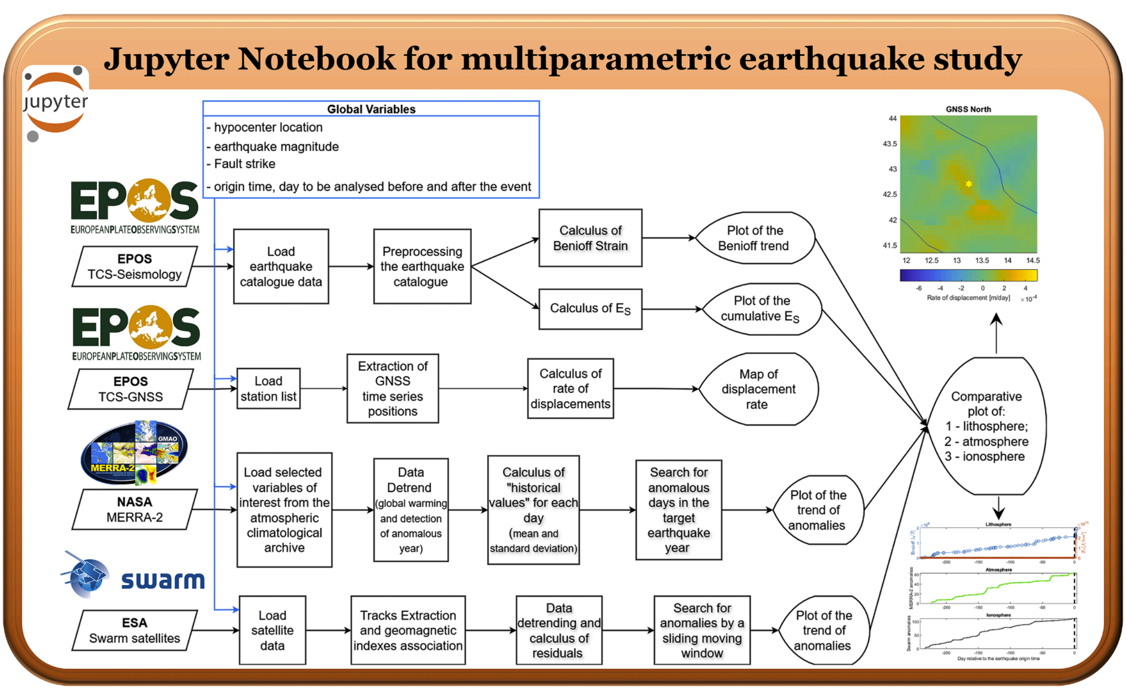

Fig. 1. General sketch of the software with workflow.

VRE_Matlab_v1.0.ipynb" or "Download_data_for_SEISMO-VRE_Python_v1.0.ipynb".

Internally, the notebook uses clearly defined configurable parameters (epicentre coordinates, origin time, magnitude, fault orientation and others) implemented as Global Variables (Fig. 2), which drive the execution of geo-layer-specific routines without modifying the Notebooks' code. Results from the three geo-layers are finally summarised in a comparative visualisation to support cross-layer interpretation.

The current implementation includes ready-to-use templates for accessing EPOS TCS Seismology and GNSS, NASA MERRA-2, and ESA Swarm data, but the architecture is intentionally modular: researchers can easily substitute or extend these modules to adapt the VRE to other data sources, observables, or natural hazard types.

The software architecture has analogies with a three tiers architecture [20] (Fig. 3) with the base layer representing the hosting infrastructure, the second tier representing the Jupyter lab environment where the code logic is executed, and the third tier is the notebook itself, where code and data analysis presentation are integrated for user purposes. The top blocks in Fig. 3 represent the data sources. Data sources can be accessed remotely in a run-time way (e.g., EPOS data), or be pre-downloaded by means of dedicated scripts (included in this work, as in the case of NASA and ESA) and or potentially include additional user data locally stored.

2.2. Software functionalities

The software implements a structured and reproducible workflow for the multiparametric analysis of earthquakes, as shown in Fig. 1. It allows users to define a set of global variables such as: (1) epicentral coordinates, (2) origin time, (3) magnitude, (4) fault strike, and (5) time window before and after the event to be analysed. The earthquake magnitude is used to compute the Dobrovolsky radius [21], which defines the spatial research area.

Data acquisition is automated through web services and supports two modes: (1) real-time data download, and (2) local file access. The current implementation includes the following data: (1) seismic catalogues from EPOS TCS Seismology, (2) GNSS time series from EPOS TCS GNSS, (3) atmospheric data from NASA MERRA-2, and (4) magnetic field data from ESA Swarm satellites.

The analysis is structured across three geophysical layers. (1) For the lithosphere, the software filters the seismic catalogue and computes Benioff strain "S" as a function of time "t" over earthquakes with magnitude "M_i" $\boldsymbol{S}(\boldsymbol{t})\left[\sqrt{\boldsymbol{J}}\right] = \sum_{i}^{i\le t}\left(\sqrt{10^{(1.5\cdot \boldsymbol{M}_i+4.8)}}\right)$ [22], the E_S parameter, which is the daily energy over the square distance "r_i" with respect to the target event $\boldsymbol{E}_S = \sum_{day}\frac{10^{(1.5\cdot \boldsymbol{M}_i+4.8)}}{r^2}$ [23], and GNSS-based displacement rates. (2) For the atmosphere, it analyses variables such as SO_2, CO, O_3, temperature, and humidity, detrends the data (by removal of polynomial), calculates historical baselines, and detects potential anomalies with a similar method defined in [24]. The degree of the polynomial is a parameter the user can set in SEISMO-VRE at the top of the atmospheric code section. We suggest selecting degree one, corresponding to a linear trend, if no particular problems are present in the historical trend. An example of variation and consequences is provided in the Supplementary Materials. (3) For the ionosphere, it extracts Swarm satellite tracks, computes residuals after baseline removal, and identifies eventual anomalous variations through a sliding window analysis along the latitude coordinate as proposed in [25]. The size of the sliding window is another parameter the user can select at the top of the ionospheric code in SEISMO-VRE. A lower window size allows for selecting localised anomalies (i.e., with a smaller size); however, it increases the number of false anomalies.

Fig. 4 presents the required inputs for all the single analysis blocks of SEISMO-VRE. In particular, the global variables are distributed and required by all the analysis blocks as they define the specific case study under investigation. In addition, each analysis segment required specific variables; for example, the lithospheric seismological analysis required the magnitude of completeness of the earthquake catalogue and the maximum depth of the events to be analysed.

Each geo-layer produces specific plots, and a final comparative visualisation summarises the trends from all three domains. This enables interpretation of potential interactions between lithosphere, atmosphere, and ionosphere, as already described in detail in [26]. A flowchart of the functionalities, main analysis steps and outputs of SEISMO-VRE is presented in Fig. 1.

The software is modular and extensible. In addition, data sources and methods can be adapted to new case studies or hazard types. Included examples, i.e., the 2016 Central Italy seismic sequence and 23 April 2025 Marmara (Turkey) earthquake, illustrate a complete use case from input configuration to final output.

3. Illustrative examples: Italian seismic sequence 2016 and Marmara-Turkey 2025 earthquake cases

In this section, we provide two illustrative examples that apply the VRE —hereafter referred to as SEISMO-VRE— to the multiparametric and multidisciplinary analysis of the preparation phase of two significant earthquakes in the Mediterranean area.

In addition, in the Supplementary Material, we provide a step-by-step guide to applying SEISMO-VRE to the volcano eruption of the Etna volcano that occurred on 3 December 2015 with a Volcanic Explosive Index (VEI) equal to 3. It was produced by the summit Voragine (VOR) crater, and it seems to be the most explosive event from 2013 to 2025 (period selected due to ESA Swarm data availability). The first part of the eruption was characterised by a rapid sequence of events, including tall lava fountains, followed in the following days by wide ash emissions [27]. Ionospheric effects induced by this volcanic eruption have been reported by Ferrara et al. [28]. However, they investigated

```
% Global variables initialization

% Earthquake origin time
EQ_year = 2016; EQ_month=8; EQ_day=24; EQ_hour=1; EQ_minute=36; EQ_second = 32;
% Epicenter latitude and longitude
epilat = 42.6980; epilon = 13.2340;
% Mainshock magnitude
EQ_mag = 6.5;
EQ_depth = 10;
Fault_strike = 153; %Here is inserted a mean strike of the two main faults: Amatrice 155 and Norcia 151
% This strike must be a positive value
day_before = 8*30;
day_after = 0;
```

Fig. 2. The figure shows the global variables used for initialising the execution of the entire notebook. Users can update these blocks both in the main Notebooks (*SEISMO-VRE_Matlab_v1.0.ipynb* and *SEISMO-VRE_Python_v1.0.ipynb*) and in the data download notebooks (*Download_data_for_SEISMO-VRE_Matlab_v1.0.ipynb* and *Download_data_for_SEISMO-VRE_Python_v1.0.ipynb*) only, and execute the multiparametric analysis in a specific spatial-temporal location associated with the target earthquake.

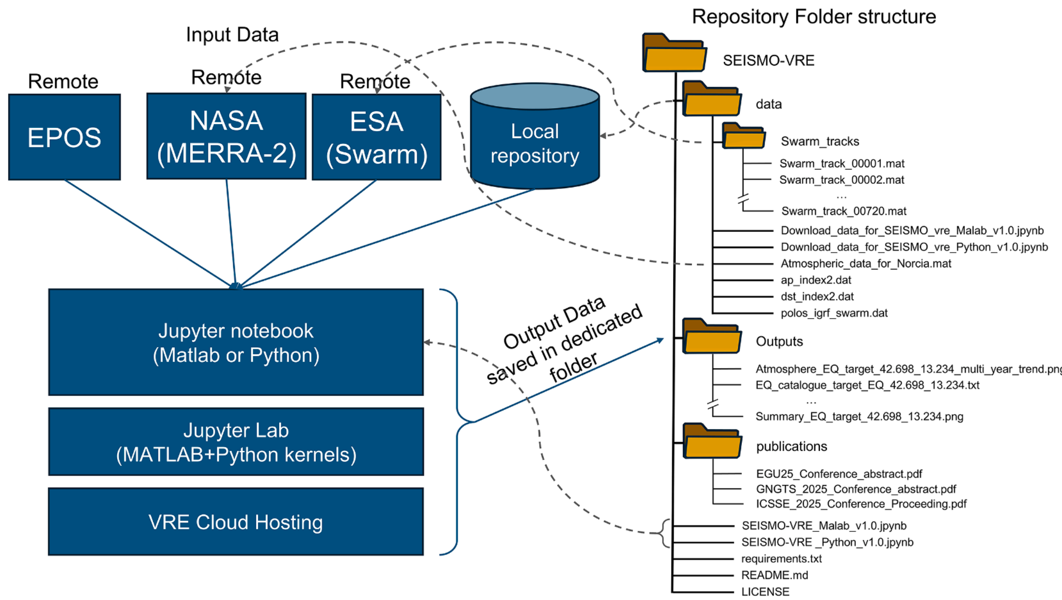

Fig. 3. Software architecture and repository structure of SEISMO-VRE. The architecture follows a three-tier logic: (1) the base layer corresponds to the hosting infrastructure, which can be a local machine or a cloud-based JupyterHub; (2) the intermediate layer corresponds to the JupyterLab environment, where MATLAB and Python kernels execute the analytical routines; and (3) the top layer consists of the Jupyter Notebook itself, where code, computation, and visualisation are integrated for user interaction. Input data (top left) are retrieved from remote research infrastructures and space agencies —EPOS (seismic and GNSS), NASA MERRA-2 (atmospheric), and ESA Swarm (ionospheric) — or from a local repository containing pre-downloaded files. Output data are automatically saved into a dedicated folder within the public SEISMO-VRE GitHub repository (right). The repository structure reflects a modular and user-friendly organisation, with separate directories for data, outputs, publications, and code, as well as standard files for documentation, requirements, and licensing.

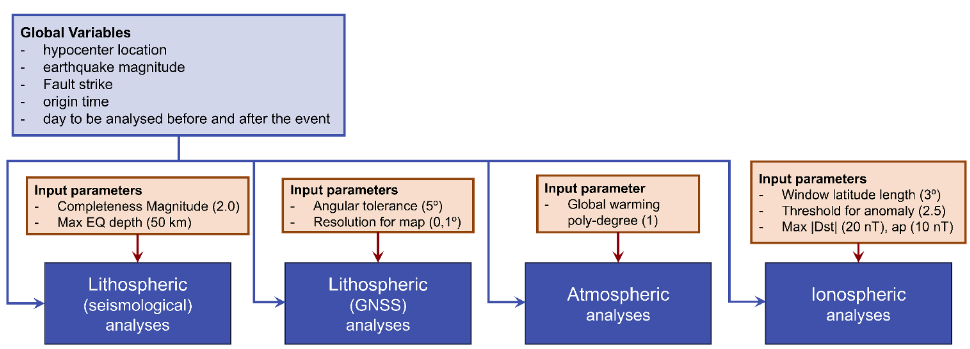

Fig. 4. Input variables and parameters required by SEISMO-VRE and each specific analysis block. The "Global variables" defined at the beginning of the Jupyter Notebook are required and used by all SEISMO-VRE analysis blocks, and they specify which case study the user is investigating. Each block has its own user-set parameters that influence the analysis results. The user can also change them and run the notebook quickly to see the differences. Default values for input parameters are provided in round brackets. Lithospheric seismological analyes has completeness magnitude of the input catalogue and maximum depth of analysis as input parameters. Lithospheric GNSS analyses have the following specific input parameters: "angular tolerance" (the maximum tolerance to consider the direction of a couple of GNSS stations with respect to the fault strike) and the map "resolution", which is the grid resolution used to interpolate the displacement data derived from the GNSS network. of the input catalogue and the maximum depth of analysis. Atmospheric analyses have as a specific input parameter the "global warming polynomial degree", which defines the degree of polynomial to be removed from the multi-year trend. Ionospheric analyses have the following specific input parameters: "Window latitude lenght" (extension of the moving window to search for anomalies along the residual track); "threshold" for anomaly detection that define if a residual signal inside the window is anomalous or not with respect to the whole track; " Max |Dst|" and "max ap", which define if the geomagnetic conditions are quiet or not as anomalies are extracted only if they are acquired within these maximum geomagnetic indices.

the Total Electron Content that SEISMO-VRE still doesn't include.

3.1. Example of SEISMO-VRE applied to the Italian seismic sequence 2016

On 24 August 2024 at 1:36 UT, a moment magnitude 6.0 earthquake struck Central Italy, near the small towns of Accumoli and Amatrice, starting a long seismic sequence that affected a wide part of Central Italy along the Apennine Mountains [29]. The largest earthquake of the entire seismic sequence occurred on 30 October 2016 at 6:40 UTC, 4 km North-East of Norcia town and with a moment magnitude of 6.5.

Regarding the parameters to run the VRE, we selected the epicentre as the average of the two main shocks, the magnitude of the largest one (i.e., 6.5), the time of occurrence of the first event, and the average of the fault directions. The fault strikes are almost aligned: 155° for Amatrice and 151° for Norcia, according to INGV solutions [30].

When executing the SEISMO-VRE with the above-mentioned parameters, we obtain the results for the lithosphere investigation shown in Fig. 5. In particular, the SEISMO-VRE produces a map of the extracted earthquake catalogue and a graph with their cumulative number over time. The selection is based on scientific constraints, including the maximum depth (to consider only the shallow seismicity), the

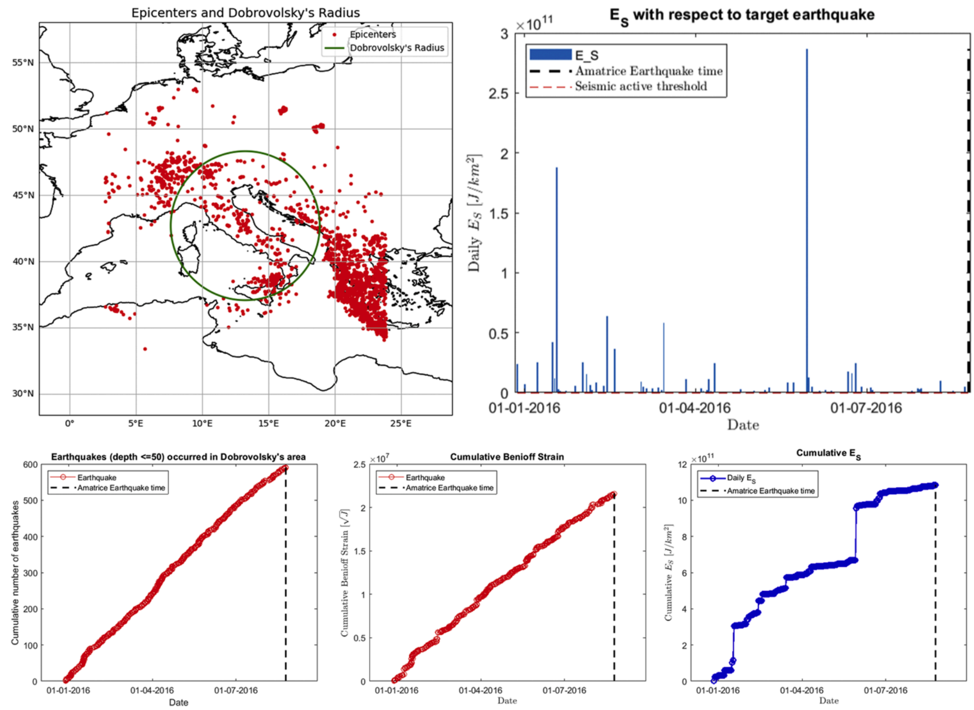

Fig. 5. Lithosphere results obtained by the SEISMO-VRE analysing the Italian Amatrice-Norcia 2016-2017 seismic sequence. The upper-left panel shows the map with longitude (horizontal axis) and latitude (vertical axis) for the earthquakes (red dots) extracted from the EPOS platform. The Upper-right panel shows the graph of the daily parameter E_S (vertical axis) over time (horizontal axis). In the lower panels: in the bottom-right, the cumulative number of earthquakes retrieved inside Dobrovolsky's circle (represented in green in the above map) and maximum depth of 50 km over time is presented; the centre-bottom panel shows their cumulative Benioff Strain over time; the right-bottom panel reports the cumulative value of E_S daily value over time. All the graphs show a vertical dashed line marking the time of the M6.0 Amatrice (Italy) earthquake on 24 August 2016.

magnitude of completeness of the catalogue, as well as only the events inside Dobrovolsky's radius, i.e., the research area where it's expected to potentially identify eventual signs of the preparation of the earthquake. In addition, the VRE computes the Benioff cumulative strain as the

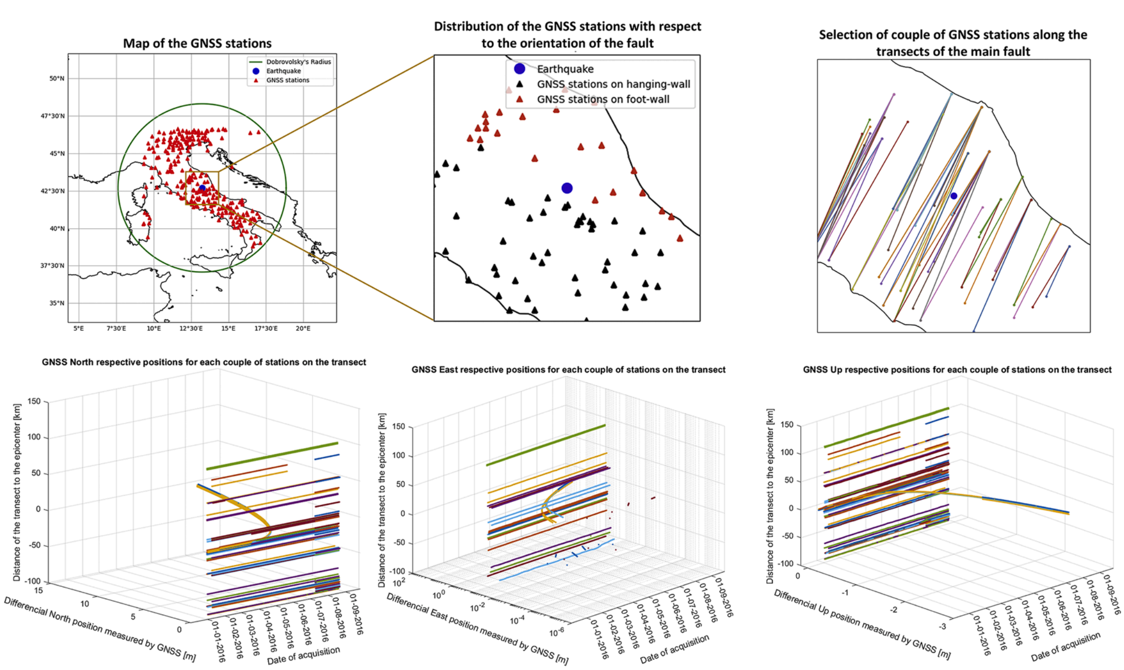

Fig. 6. GNSS investigation along the transects of GNSS couple of stations for the North, East and Up components. In the upper-left panel, the map of the available GNSS stations inside the EPOS is presented as red triangles. The blue circle represented the position of the target earthquake, and the green circle is Dobrovolsky's circle. In the top-centre map, the selected stations closer to the epicentre are reported and divided into those on the footwall as red triangles and those on the hanging wall as black triangles. The top-right figure shows all the possible couples of the previous stations (one red and one black) on the transect direction of the main fault. For each of them, their relative positions over time are reported in the bottom three graphs for North, East and Upward directions, respectively. These graphs report on the x-axis the time, on the y-axis the displacement, and on the z-axis (vertical) the distance of the transect with respect to the epicentre.

square root of the released energy by each earthquake [22] and E_S parameter in the form of daily value and their cumulative trend. E_S was defined as the daily released earthquake energy weighted for square distance from a geographical point [23,31]. In the original article, the distance is calculated from the measurement station, while here it's from the target earthquake. These calculations aim to identify an eventual seismic acceleration and or activation before the target event.

In order to investigate the tectonic displacement rates, the SEISMO-VRE retrieves GNSS data and, in particular, the list of stations from EPOS GNSS services (see Fig. 6). Then, it selects the closer ones to the target earthquake. For this purpose, it estimates the length of the broken fault using the equation provided by Doglioni et al. [32], and selects stations within ten times that length of the broken fault. The algorithm then divided the stations according to their position with respect to the strike of the main fault, i.e., dividing them into those on the footwall and those on the hanging wall. Using these two groups, it identifies couples of stations whose direction is perpendicular to the main fault strike, i.e., they are along a transect of the main fault with a tolerance (e.g., 5°). For each couple of GNSS stations, SEISMO-VRE calculates the relative displacement as a function of time in the North, East and Up components, plotting it in a three-dimensional graph, which provides information of the distance of the transect from the target earthquake on the vertical axis. The scope of this analysis is to identify a possible variation in the tectonic movements due to lock or acceleration of the fault. Considering that a normal fault moves in opposite directions, these variations along the transects were investigated. For other focal mechanisms (especially strike-slip), this section must be revised to reflect the fault's preferred slip direction.

In addition, the SEISMO-VRE estimates the rate of variations of GNSS displacements recorded in the research area (in this case, Central Italy) every week. For this purpose, the GNSS data are first homogenised and linearly interpolated station-by-station wherever possible. After a triangular linear spatial interpolation with the method described in [33] over a regular grid with a default resolution of 0.1° is performed using the available stations, the maps are produced. One of them is shown in Fig. 7. The advantage of this second approach is that all available GNSS stations are combined in order to obtain more accurate information and possibly isolate local effects related to fault activity.

The analysis of the atmosphere aims to identify possible physical or chemical alterations of this geo-layer induced by seismic fault degassing or air ionisation. Fig. 8 shows the output from the SEISMO-VRE analysing the atmospheric climatological data. To account for the particular trend of carbon monoxide over the years, a 5th-degree polynomial fit was chosen. A similar approach was applied to the Mw 6.8 Lushan (China) 2013 earthquake for the same reason [34]. This is a different selection applied to the same case in a preliminary investigation we presented in a previous conference paper, where we used a linear trend [35]. The anomalous parameters over time are also represented, along with their cumulative values. The CO does not report any anomalies even with the previous multi-year linear detrend. The reason for the lack of CO anomalies could be due to very low values recorded in 2016 in Central Italy, but we do not exclude that a more sophisticated technique (for example, the one proposed by Zhang et al. [36]) could be able to extract possible CO anomalies that are supported by measurements of linked geochemical observables [37]. The absence of CO anomalies detected by SEISMO-VRE could also be due to the low resolution of the original data if the anomalous zone was very confined in a few kilometres around the fault.

For ionospheric analysis, the SEISMO-VRE collects all Swarm satellite data acquired within the research timeframe. The goal of this analysis is to extract ionospheric disturbances to determine whether they could be induced by seismic activity. Then it selects only the tracks that crossed the area of interest and estimates the residuals, i.e., it removes the baseline to search for possible small variations of the signal along the track. The technique was already applied to single earthquakes as well as statistical investigations [2,25,38–40], and it's based on an estimation of the first numerical derivative of the signal and the subtraction of a spline from the data. Fig. 9 presents ten randomly chosen examples of residuals of the Y-East component of the magnetic field acquired by Swarm satellites.

In analogy to the other investigations, SEISMO-VRE extracts anomalous signals using a sliding window from the residual acquired in geomagnetic quiet conditions and inside the research area. The use of the window compared with the full track is based on the assumption that eventual seismo-ionospheric signals are expected to be localised in the research area and not diffused on the whole track (that they are more likely global ionospheric disturbances). Finally, it produces a summary graph of lithosphere, atmosphere and ionosphere investigations that we present later, compared to the next example of application of SEISMO-VRE to another earthquake.

3.2. Example of SEISMO-VRE applied to Marmara-Turkey 2025

On 23 April 2025 at 9:49:11 UTC, an earthquake of moment magnitude Mw = 6.2 hit the Marmara region in Turkey about 67 km west-southwest of Istanbul. According to USGS analyses, the earthquake's rupture was a normal focal mechanism with a slight oblique component. The epicentre was entirely localised in the Sea of Marmara, and the corresponding hypocentre was at a depth of 12.7 km. Consequently, this earthquake can be considered a case study with several

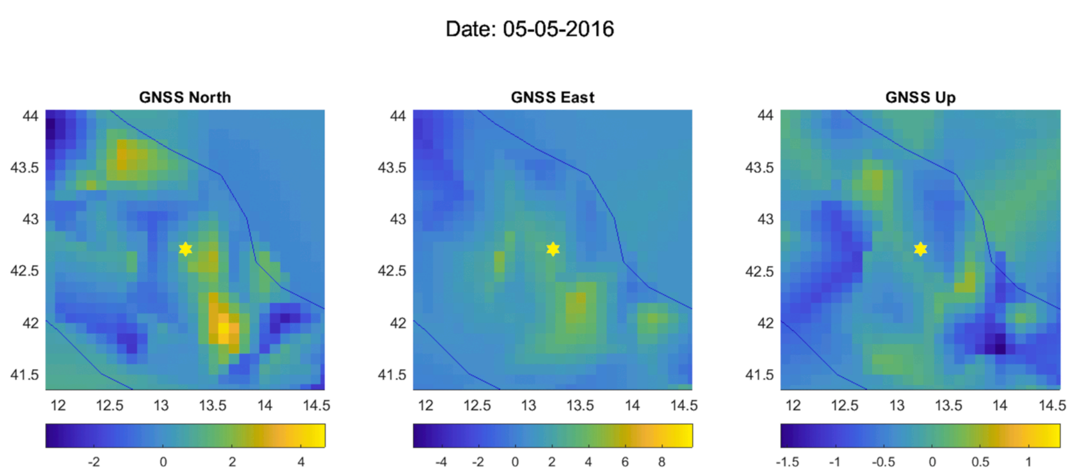

Fig. 7. Maps (latitude vertical axis and longitude on horizontal axis) of the weekly rate of Central Italy GNSS displacement in the North, East and Up components estimated on 5 May 2016. The epicentre of the target event of SEISMO-VRE is represented as a yellow star, and blue lines are the coastlines.

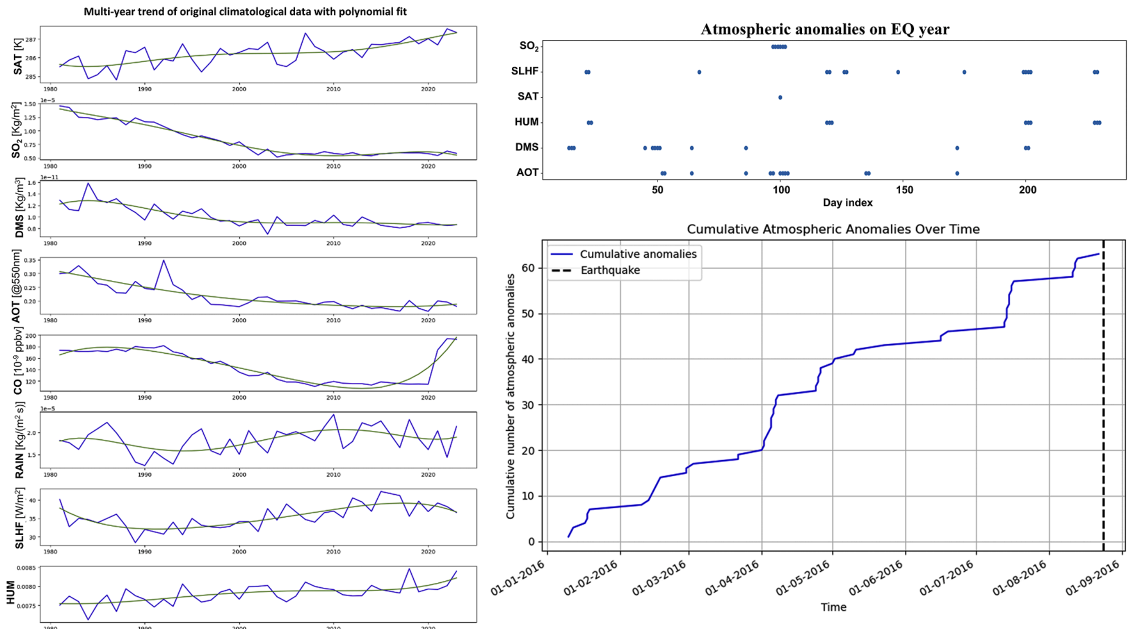

Fig. 8. Results of atmospheric climatological data investigation by the multiparametric earthquake VRE applied to the Italian Seismic sequence 2016-2017. On the left panel, the multi-year trends are reported as blue lines for (from top to bottom) surface air temperature, sulphur dioxide, dimethyl sulphide, Aerosol Optical thickness, carbon monoxide, total precipitation, surface latent heat flux and relative humidity parameters, respectively. The "global warming" polynomial fit is overplotted as a green line for each parameter. In the upper-left panel, the extracted anomalies (SO_2, SLHF, SAT, HUM, DMS, and AOT) over time are reported as blue dots, and their cumulative time series are presented in the bottom-right panel. The vertical dashed line marks the time of the M6.0 Amatrice (Italy) earthquake on 24 August 2016.

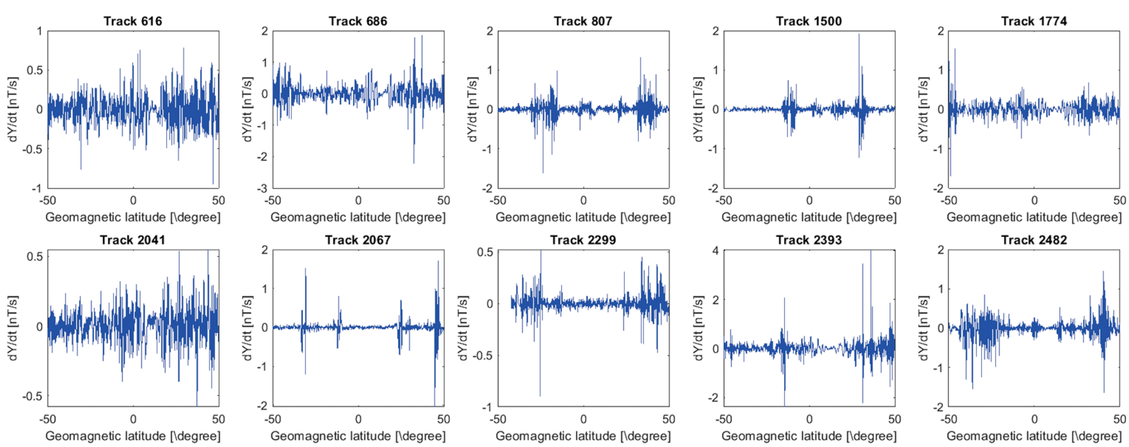

Fig. 9. Examples of 10 tracks (randomly selected by SEISMO-VRE) of residuals of the magnetic field Y-East component data acquired by ESA Swarm satellites. The title of each graph reports the track number. The vertical axis is the residual magnetic signal in the Y (East) component of the magnetic field. The horizontal axis is the geomagnetic latitude. Some tracks exhibit a distributed signal along their length (e.g., 616, 2041); others are more localised and often at magnetic conjugates. These residuals from SEISMO-VRE plot these tracks to check the residual magnetic data, which are used to extract ionospheric anomalies using a sliding window, ensuring that their centres fall within the research area and occurred in geomagnetic quiet conditions.

analogies to the previous one (e.g., similar magnitude, focal mechanism, depth) but also some differences (e.g., the location in the sea, the more active tectonic setting of Turkey compared to the central Apennines of Italy).

Considering the above factors, we decided to run SEISMO-VRE using the data from the 2025 Turkey Marmara earthquake for a comparison. We set the hypocentral coordinates, origin time, and magnitude in the global variable, and we applied the same time interval as the previous case studies, i.e., 8 months before the earthquake and no time after the event (0 days). The lithospheric investigation produced the analyses shown in Fig. 10. We noted a higher number of earthquakes, as expected for this very active seismic area, for the plate boundary between the African and Eurasian plates. Furthermore, the epicentres are concentrated in the lower part of the region, i.e., on the main plate boundary. The results show a significant difference from a seismological point of view, i.e., the presence of a foreshock about 30 minutes before the target earthquake, well depicted by the E_S parameter. On the other hand, in the previous case study, the M6.0 Amatrice 2016 earthquake occurred suddenly without any foreshock or clear pre-seismic activity. The scientific reasons for the presence or absence of foreshocks, as well as for seismic acceleration before the mainshock, remain debated. For example, about 7 years before the M6.3 L'Aquila earthquake on 6 April 2009, there were foreshocks and seismic acceleration [41]. The event occurred in a very close region, with the same focal mechanism, depth, magnitude, and geological context. Still, the 2016 Italian Seismic sequence was preceded by a seismic quiescence, which could be a

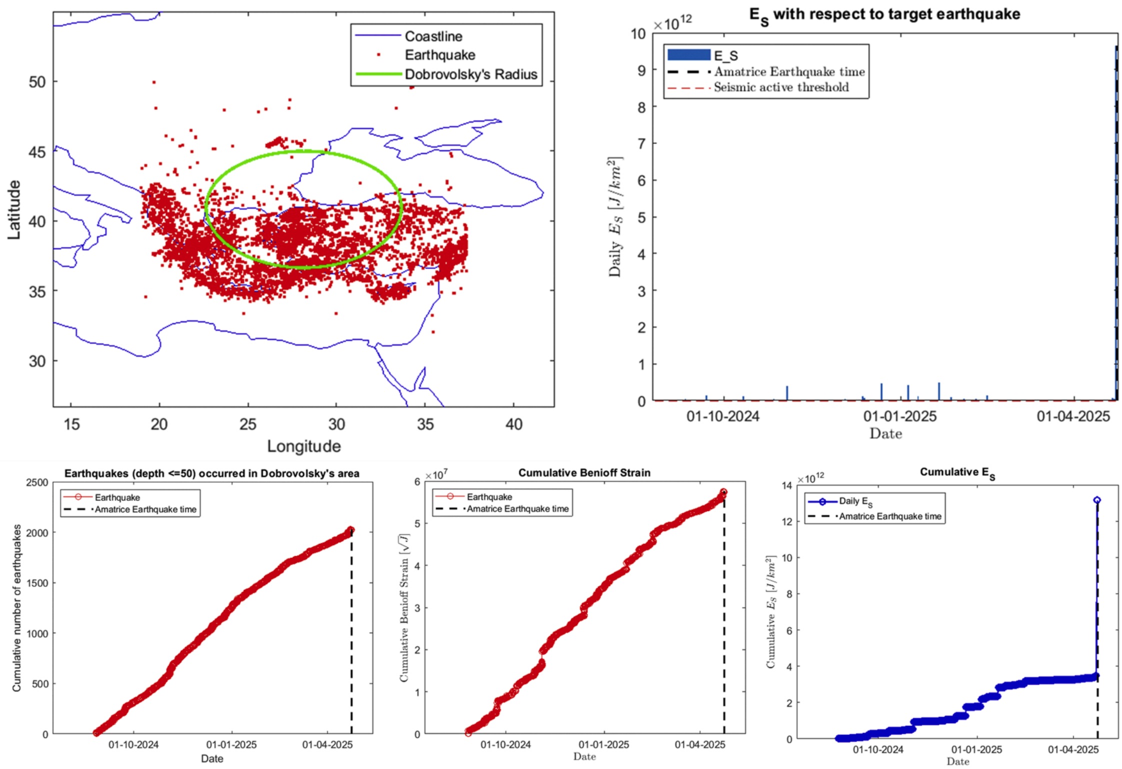

Fig. 10. Lithosphere results obtained by SEISMO-VRE analysing the Mw = 6.2 Marmara-Turkey 23 April 2025 earthquake. The maps and the panels represent the same quantities in the same axes of Fig. 4. In this case, the map shows more earthquakes (especially in the Southern side) due to a higher local (Turkey-Greece) seismic rate compared to Central Italy. The ES parameter and its cumulative (right side graphs) is predominated by the foreshock that is associated with a very high value of E_S parameter, even though not so high-magnitude (but very close to the incoming earthquake) as in the Benioff cumulative graph (bottom-centre panel), no particular increase is associated with the same event.

complementary seismological sign of an incoming significant earthquake [42].

SEISMO-VRE processed the atmospheric and ionospheric data in the same manner as the previous case study. It reconstructs the same analysis applied to this other research area and in the eight months before this other earthquake. Finally, the summaries of both analyses are presented in Fig. 11. The figure shows the summary from the first case study on the left and that of the second on the right. The summary serves as a basis for searching for possible Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) before a seismic event, following a similar approach applied to different case studies in recent years [2,26,43,44]. One of the advantages of adopting SEISMO-VRE is to compare different case studies using exactly the same data and applying the same method, as shown in Fig. 11. Furthermore, SEISMO-VRE permits discussing similarities and differences among multiple seismic events, comparing the trends from different geo-layers.

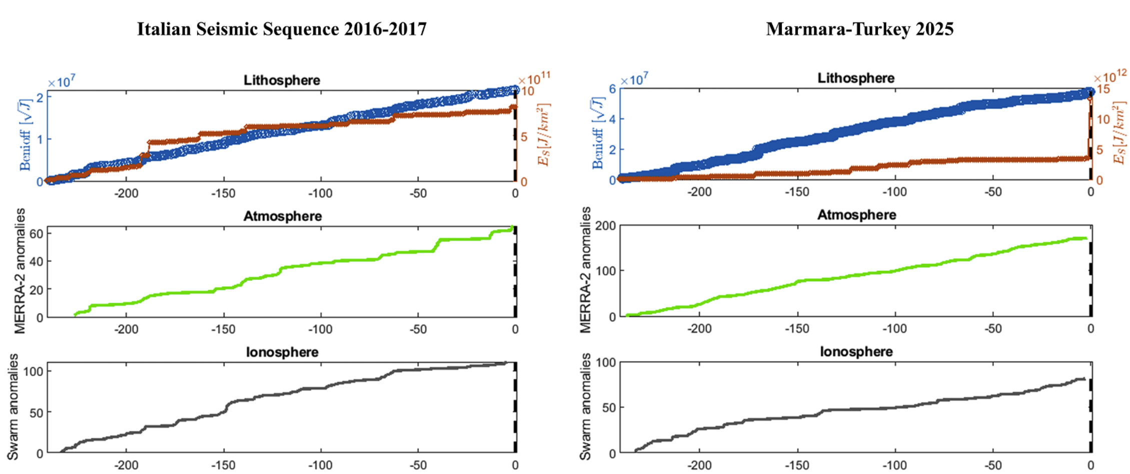

Fig. 11. Summary graph output of the SEISMO-VRE applied to the Italian Seismic Sequence 2016-2017 on the left and on the Marmara-Turkey 2025 on the right. For both the right and left sides, the scientific analyses (data and methods) are exactly the same. The upper graph presents the lithospheric analysis with a blue curve, which is the Benioff cumulative strain, and a brown curve, which is the cumulative E_S daily parameter. In the middle panel, the green line presents the cumulative number of atmospheric anomalies extracted from MERRA-2 data. In the bottom graph, the grey line presents the cumulative number of Swarm ionospheric magnetic field anomalies in the research area.

4. Impact

The SEISMO-VRE software contributes to advancing geoscientific research by enabling reproducible, modular, and multidisciplinary investigations of earthquake preparation phases. It provides a valuable tool to discriminate among the different Earth geo-layer coupling theories [45–49] by the timing and comparison of the anomalous parameters in the lithosphere, atmosphere and ionosphere, contributing to understanding the physical-chemical mechanism of Earth's system layers interactions.

Its architecture supports the integration of heterogeneous data sources across multiple geophysical domains—seismology, GNSS, atmospheric science, and ionospheric physics—within a single analytical environment. Thanks to the modularity of SEISMO-VRE, the user community can easily integrate other earthquake-related investigations, such as analysis of soil liquefaction [50]. For this purpose, new specific techniques of data exploitation can be integrated, for example, the multivariate adaptive regression splines [51] or optimised algorithms to determine the probability of soil liquefaction [52]. In addition, the tool can be expanded to other disciplines such as geological engineering for the estimation of the seismic wave on anthropogenic buildings and infrastructures [53]. Furthermore, recent multisource data fusion methods have successfully shown to be helpful in the geotechnical field modelling [54]; consequently, the application of SEISMO-VRE in this field has a potentially great impact.

SEISMO-VRE supports FAIR data reuse [55], by integrating validated datasets from major infrastructures and agencies like EPOS, NASA, and ESA, thus promoting shared scientific practices, reducing technical barriers to cross-domain research, and facilitating deployment on collaborative platforms such as JupyterHub.

Beyond earthquake-related applications, the modular design of SEISMO-VRE facilitates its reuse in a variety of natural and anthropogenic hazard contexts, such as volcanic eruptions, industrial accidents, or other anthropogenic hazards. For example, similar methodologies have already been applied to the 2019 La Palma volcanic eruption [56], the 2020 Beirut explosion [57], and the 2011 Fukushima nuclear disaster [58], all of which showed signs of lithosphere–atmosphere–ionosphere interactions.

From an operational perspective, the software architecture allows for the implementation of automated workflows triggered by real-time seismic events (e.g., $M \geq 5.5$ and maximum depth of 50 km as done in [40]) within specific geographic areas. In this sense, SEISMO-VRE could be considered as a potential asset for early-stage anomaly extraction from geophysical observations in a posteriori earthquake scenario. We would note that "*Early-stage assessment*" must not be confused with "*Early warning systems*", as SEISMO-VRE is not designed to predict any effects but only for analyses of the available data. Limitations of this approach is the availability of the original data. The most critical is the atmosphere data, which are updated monthly and released about halfway through the next month, so they could introduce an average delay of 1.5 months. On the other hand, the earthquake catalogue is updated in a time of the order of minutes, and Swarm magnetic data are updated with a delay of about 3 days, but the "FAST" product, released as soon as the data are downloaded and processed from satellite, is available with a delay of less than one day.

Finally, SEISMO-VRE demonstrates how open scientific software can foster interoperability between infrastructures, data reuse, and transparent workflows for multidisciplinary hazard analysis.

5. Conclusions

This paper presents SEISMO-VRE, an open-source Virtual Research Environment designed to support the multiparametric and multidisciplinary analysis of the earthquake preparation phase. Built as a Jupyter Notebook running with both MATLAB or Python kernels, the tool provides a complete, automated pipeline from data acquisition to comparative visualisation across three geophysical domains: lithosphere, atmosphere, and ionosphere.

The software retrieves data from authoritative organisations such as EPOS, NASA, and ESA in a flexible way (run-time access, pre-download, or local) according to the global variables set by users, matching the earthquake that they want to investigate.

Thanks to its modular design, users can easily extend or adapt the tool to new case studies, with additional parameters, different hazard types and alternative or additional data sources.

Therefore, SEISMO-VRE reduces the technical burden of connecting to different data sources, integrating and processing heterogeneous geophysical datasets, thereby empowering researchers to focus on scientific interpretation and comparison of the results. This has been demonstrated by applying the software to two case studies: one related to the Italian seismic sequence 2016 and the other to the $Mw = 6.2$ Marmara (Turkey) 23 April 2025 earthquake.

SEISMO-VRE implementation with Jupyter notebooks facilitates reproducibility, fosters cross-domain collaboration, and supports the broader adoption of open science practices in Earth system research.

Future developments may include real-time operational deployment, expansion to additional data services, and integration into cloud-based research platforms. As such, SEISMO-VRE offers a robust, flexible, and extensible foundation for advancing the study of complex geohazards in a reproducible and collaborative way.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, generative AI (ChatGPT, version 4) was used to assist in the drafting of selected sections related to software architecture and to improve the overall clarity and fluency of the text. The authors reviewed, revised, and edited all AI-generated content to ensure accuracy and originality, and take full responsibility for the final version of the published work.

SEISMO-VRE software repository is available with the following DOI: https://doi.org/10.5281/zenodo.17602394.

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CRediT authorship contribution statement

Dedalo Marchetti: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daniele Bailo:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Giuseppe Falcone:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Jan Michalek:** Writing – review & editing, Validation, Supervision, Resources, Data curation. **Rossana Paciello:** Writing – review & editing, Validation, Resources, Funding acquisition, Data curation. **Alessandro Piscini:** Writing – review & editing, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.softx.2026.102538.

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